

CS 486/586 – Homework 1

Student 1 name and ODIN id: _____

Student 2 name and ODIN id: _____

Please read this entire assignment carefully before you begin. Many questions build on earlier ones, and missing small details may cause later answers to be incorrect.

Submission Instructions

- Since this is a group assignment, I would like to clearly understand each member's contribution. Please include a table on the first page listing all group members and the sections they completed (e.g., Esme – Answers 1, 2, and 3).
- Each member must submit their own copy of the homework. All submissions should be identical across group members. If your submission does not match your group's work, please ensure it is aligned prior to submission; otherwise, the lowest score among the group submissions may be applied.
- Edit this document directly and place your answers immediately below each question.
- Answers must appear in the same order as the questions.
- If your answers are out of order or hard to locate, they may be marked incorrect so put them in the same order as the question.

Advice: If you are unable to complete an assignment on time, submit whatever you have completed by the deadline. You will receive full credit for the portion you submit. The remaining incomplete portion may be eligible for up to 50% partial credit if completed and submitted later, and it will be graded at the end of the term. This opportunity is intended for partial recovery and does not replace timely submission.

Assignment deadlines are firm, and extensions will not be granted. Please do not email requesting deadline extensions. In case of exceptional circumstances or emergencies, you are encouraged to discuss your situation during office hours.

Accommodations

If a group member has an approved accommodation, you are still expected to collaborate and aim to complete the assignment by the deadline as a group. If accommodations are needed, please discuss them with me in person. Accommodations will apply only to the individual student's portion of the work. Any incomplete portions from other group members will still be subject to grading. To avoid issues, I strongly encourage groups to plan ahead and complete the assignment on time together.

Screenshots

All questions that require screenshots must include:

- Column names
- The required number of rows asked in the question
- The visible query result pane (not cropped values only)

Technical Requirements

- **Database:** PostgreSQL
- **Tools:** DBeaver, PostgreSQL CLI ([psql](#)) recommended
- Queries written for other SQL dialects (MySQL, SQLite, etc.) may not receive credit.

Dataset

This assignment uses a **sample Yelp dataset** of donut shops in the United States (2023).

- The data is provided **only for this course**.
- Do **not** reuse or redistribute the dataset.

Point Breakdown

- Each question is worth **5 points**.
- Questions **1 and 2** are free points if completed correctly.

Questions

1. Schema Creation

Run the SQL script **Lab1-tables.sql** to create the schema and table.

No screenshot required.

2. Data Loading

Run the SQL script **Lab1-data.sql** to load 5,000 rows of data.

Important notes:

- Loading via DBeaver may take ~50 seconds off-campus.
- DataGrip may hang or fail.
- Loading via PostgreSQL CLI (**psql -f**), you'll find this file as part of the assignment in Canvas and on ada/babbage in `~rchaney/Classes/cs486/Labs/Lab1`:

```
PSQL -f Lab1-tables.sql
PSQL -f Lab1-data.sql
```

No screenshot required.

3. Data Validation

Validate that exactly **5,000 rows** were loaded.

```
SELECT COUNT(*) FROM yelp_donut.donut_data;
```

- Include your SQL query.
- Include a screenshot of the result.

4. First 10 Rows

Write a SQL query that displays the **first 10 rows** of the table.

Requirements:

- Select **all columns** using *****
- Use **LIMIT**
- Do **not** use **ORDER BY**
- Include your SQL query.
- Include a screenshot of the 10 rows.

5. Ordering by County

Modify Question 4 to **sort the data by county**.

- Include your SQL query.
- Include a screenshot of the first 10 rows.
- Answer the following:

What seems odd about the first row of the result set, given how counties are usually named or ordered?

6. OFFSET

Modify Question 5 to **skip the first 1,000 rows** using **OFFSET**.

- Include your SQL query.
- Include a screenshot of the 10 rows returned.
- Answer the following using this format:

City: ____

County: ____

Where is A J's Donuts located?

Questions 7–13

For **Questions 7–13**, return **only the following columns**:

name, city, county, state, country, rating, review_count, price_indicator

7. Donut Shops in Oregon

Write a SQL query that returns donut shops located in **Oregon**.

- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

8. Most Reviews (Oregon)

Which donut shops in Oregon have the **most reviews**?

- Sort by **review_count**
- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

9. Fewest Reviews (Oregon)

Modify Question 8 to show donut shops in Oregon with the **fewest reviews**.

- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

10. At Least 100 Reviews

Modify Question 8 to show donut shops in Oregon with **at least 100 reviews**.

- Include your SQL query.
- Include a screenshot of the **first 5 rows**

11. Highest Rating (Filtered)

Modify Question 10 to show donut shops in Oregon with the **highest rating**.

- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

12. Portland (Any State)

Write a SQL query that shows donut shops located in a city named **Portland**.

- Do **not** order the data.
- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

13. Portland, Oregon

Modify Question 12 to show donut shops located in **Portland, Oregon**.

- Do **not** order the data.
- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

Questions 14–16: Column Changes

14. Distinct States

Write a SQL query that returns the **distinct states** that have donut shops.

- Order by state.
- Include your SQL query.

- Include a screenshot of the **first 5 rows**.

15. Portland States

Modify Question 14 to show distinct states that have a donut shop in a city named **Portland**.

- Include your SQL query.
- Include a screenshot of the results.

16. Vegan-Friendly Shops

Write a SQL query that returns:

`name, city, liked_by_vegans`

Only include donut shops in **Portland, Oregon** that are liked by vegans.

- Include your SQL query.
- Include a screenshot of the results.

Final Queries

Return to using the following columns:

`name, city, county, state, country, rating, review_count, price_indicator`

17. Highest Rated in Oregon

Which donut shop in Oregon has the **highest rating**?

- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

18. Perfect Ratings Nationwide

Modify Question 17 to show **all donut shops nationwide** with a rating of **5**.

- Order by state and city.
- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

19. Count of Perfect Ratings

Modify Question 18 to show **how many donut shops** have a rating of **5**.

- Use the **COUNT()** aggregate function.
- Include your SQL query.
- Include a screenshot of the result.

20. Names Beginning with 'Co'

Modify Question 17 to show donut shops in **California** whose name begins with '**Co**'.

- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

21. Lowercase 'co'

Modify Question 20 to show donut shops in California whose name begins with '**co**'.

- Include your SQL query.
- Include a screenshot of the results.

22. Any Capitalization

Modify Question 21 to show donut shops in California whose name begins with '**co**' in **any capitalization**.

- Include your SQL query.
- Include a screenshot of the **first 5 rows**.

23. Donut in the Name

Write a SQL query that returns donut shops in **Oregon** that contain the word '**donut**' (any capitalization) anywhere in the name.

- Include your SQL query.
- Include a screenshot of the results.